Robot Motion: review

Probabilistic Robotics, chapter 5

Lesson t2.4 · Thrun, Burgard, Fox

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Review map

1. 5.2 Motion as a distribution
2. 5.3 Velocity model, Tables 5.1 to 5.4
3. 5.4 Odometry model, Tables 5.5 and 5.6
4. 5.5 Motion and maps, Table 5.7
5. Cards C1 to C5, symptoms, mistakes

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5.2 Motion is the predict step

$$p(x_t \mid u_t, x_{t-1}) \tag{5.3}$$

- p. 118: overestimating uncertainty is more robust; p. 121: velocity for planning, odometry for estimation
- Trap: $\theta \in S^1$, so every angle difference is wrapped

Trap: every angular innovation is wrapped before it meets a gain

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5.3 Exact motion on a circle

$$x' = x - \frac{v}{\omega}\sin\theta + \frac{v}{\omega}\sin(\theta + \omega\Delta t) \tag{5.9}$$

- p. 127; agrees with RK4 of the unicycle to 8×10^{-13}
- Trap: ω = 0 puts the centre at infinity; chord form avoids the division

Trap: ω = 0 divides by zero

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5.3.3 Final rotation

$$\theta' = \theta + \hat{\omega}\Delta t + \hat{\gamma}\Delta t \tag{5.14}$$

$$\hat{\gamma} = \varepsilon_{\alpha_5 v^2 + \alpha_6 \omega^2} \tag{5.15}$$

- p. 129; two noise variables leave a 2-D sheet in pose space
- Trap: keep $\hat{\gamma}$ even when you think $\alpha_5 = \alpha_6 = 0$

Trap: without $\hat{\gamma}$ the density is zero off a 2-D sheet

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5.3.1 Table 5.1: the density

- pp. 121–123: invert (5.9), score the three errors independently (5.32)
- **$\hat{v} = (\Delta\theta/\Delta t)\, r$** $*$ on line 7 loses the sign of the motion
- Line 2 is 0/0 for a straight move; lines 6 and 9 need wrapping
- (5.32) is a density over $(\hat{v}, \hat{\omega}, \hat{\gamma})$, not over poses
- Fix: chord form `velocity_inverse`, or λ patch from (5.17)

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5.3.2 Tables 5.2 and 5.4: error samplers

- Both samplers have variance b^2; the triangular one stops at $\pm\sqrt{6}\, b$
- Print error p. 132: the 12-uniform support is $[-6b, 6b]$, not $[-2b, 2b]$
- Variance checks: 1.0003 and 1.0007 over 10^6 draws
- Trap: $b^2 = 0$ at rest makes the normal density 0/0; floor b^2

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5.4 Odometry decomposition

$$\delta_{\text{rot1}} = \text{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta} \tag{5.34}$$

$$\delta_{\text{rot2}} = \bar{\theta}' - \bar{\theta} - \delta_{\text{rot1}} \tag{5.36}$$

- pp. 132–135; Table 5.5. Trap: wrap δrot2 − δ̂rot2 (p. 135), and every line like it
- Trap: δtrans = 0 makes atan2(0, 0) arbitrary

Trap: unwrapped δ differences lose density at ±180°

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5.4.2 Odometry sampler

$$x' = x + \hat{\delta}_{\text{trans}} \cos(\theta + \hat{\delta}_{\text{rot1}}) \tag{5.40}$$

- pp. 136–138; Table 5.6 uses the measured δ in the variances
- Trap: dead reckoning grows faster than linearly: 0.118 m at 3 m, 1.894 m at 17.0 m

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5.5 Motion and maps

$$p(x_t \mid u_t, x_{t-1}, m) = \eta \frac{p(x_t \mid u_t, x_{t-1})\, p(x_t \mid m)}{p(x_t)} \tag{5.48}$$

- pp. 140–143; Table 5.7 has two lines numbered 3
- Trap: end-pose check tunnels through walls; p. 141 warns about termination

Trap: one end-pose check lets particles through walls

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C1 Densities and samplers: the tricks

- **Log density, summed**: the naive density is exactly 0.0 beyond 38.6 b
- **Floor on b^2** in each term's units: at rest the naive density is 0/0
- **Unit normals times $\sqrt{b^2}$**, not normal(0, b): works for $b^2 = 0$ or an array
- Triangular density is zero beyond $\pm\sqrt{6}$ b: mix in a small uniform term

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C2 Velocity forward, chord form

```python
def velocity_forward(X, v_h, w_h, g_h, dt):
    X = np.asarray(X, dtype=float)
    half = 0.5 * w_h * dt
    s = v_h * dt * np.sinc(half / np.pi)
    th = X[..., 2]
    return np.stack([
        X[..., 0] + s * np.cos(th + half),
        X[..., 1] + s * np.sin(th + half),
        wrap(th + (w_h + g_h) * dt)], axis=-1)
```

- Nothing divides by $\hat{\omega}$, so no straight-line branch
- Wrap θ'; one call for all M particles

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C3 Velocity inverse: book vs chord form

Table 5.1 as printed

- Line 7: $\hat{v} = (\Delta\theta/\Delta t)\, r_*$ loses the sign
- Line 2 is 0/0 for a straight move
- Lines 6 and 9 need wrapping

Chord form (C3)

- Displacement along and left of heading, then atan2
- Sign of $\hat{v}$ from displacement along heading
- One angle difference left, wrapped
- Valid for less than a half turn per step

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Pose density: divide by the Jacobian determinant, floor $|\hat{v}|\Delta t$ at 1 cm

C4 Odometry parameters and variances

```
def odometry_params(P0, P1):
    dx, dy = P1[..., 0] - P0[..., 0], P1[..., 1] - P0[..., 1]
    trans = np.hypot(dx, dy)
    rot1 = np.where(trans < TRANS_EPS, 0.0,
                    wrap(np.arctan2(dy, dx) - P0[..., 2]))
    rot2 = wrap(P1[..., 2] - P0[..., 2] - rot1)
    return rot1, trans, rot2

def rot_for_noise(rot):
    return np.minimum(np.abs(rot), np.abs(wrap(rot - np.pi)))
```

- Rotation in place: δrot1 = 0 below 1 cm
- Reversing is not a half turn: measure from the nearer of ahead and back

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C4 Odometry sampler per step

1 Once per step

Lines 2–4 depend only on u: compute δ once, not per particle

2 Perturb

Measured δ in the variances, as (5.44)–(5.46) do

3 Move

Read θ before overwriting it; each particle uses its own heading

- Keep the last odometry pose used, not a sum of 20 ms δs

Density and sampler must describe one distribution

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C5 Map-based sampler with a retry cap

```
def sample_motion_model_with_map(u, X, sampler, in_free, rng,
                                 max_tries=10):
    Xn = sampler(u, X, rng)
    ok = np.asarray(in_free(Xn), dtype=bool)
    for _ in range(max_tries - 1):
        if ok.all():
            break
        bad = ~ok
        Xn[bad] = sampler(u, X[bad], rng)
        ok[bad] = in_free(Xn[bad])
    return Xn, ok.astype(float)
```

- Redraw only the failures; π = 0 where every draw failed
- Configuration space grown once, at map load

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Density vs sampler: total variation

0.086

velocity, as printed

0.044

velocity, with Jacobian

0.153

odometry, as printed

0.064

odometry, Jacobian and measured δ

- Floors from two sample sets: 0.063 velocity, 0.091 odometry

Dividing out the Jacobian and using measured δ closes the gap

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Cards: shapes and cost

Card	Shapes	Cost
C1 densities, samplers	a, b^2: scalar or (M,)	O(1) per call
C2 velocity sampler	X (M, 3); u = (v, ω)	O(M) per step
C3 velocity density	Xp (N, 3); X (3,)	O(1) per hypothesis
C4 odometry	u two (3,) poses; X (M, 3)	O(M) sample, O(1) density
C5 with map	X (M, 3); π (M,)	1/acceptance draws, capped

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Symptom to fix

Symptom	Cause	Fix
Confident and wrong	α too small	Inflate α; fit α (T5)
Density zero off the sheet	$\alpha_5 = \alpha_6 = 0$	Noise (5.15)
NaN at rest	$b^2 = 0$	Floor on b^2
NaN on straight lines	$\omega = 0$	Chord form
Right turns unlikely	Line 7 sign	Sign from displacement
Likelihood collapse at ±180°	Unwrapped difference	Wrap
Particles behind wall	End-pose check	Shorter steps
Density and sampler differ	Sign, wrap or variance	TV test of L9

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Mistakes already made (REVIEW.md)

Where	Error	Fix
L6 trap 3	Cards said to keep δ variances	Cards use measured δ; book disagrees with itself
T2.5(c)	Figure for exercise 4 does not exist	Point to text on p. 146
T2.4(c)	Variance 'linear' in path length	Double integrator T^3; chapter models via heading
L1 §5.2.1	Book and Barfoot differ on chaining	Both are SE(2) composition; wrap θ
C4 trick 5	x–y block 'a rotation scaled'	Columns orthogonal, norms δtrans and 1
L9	'Jacobian 1/δtrans'	Determinant is δtrans
Tracked robot	Odometry 'the worse source'	Untested on the hull; see next slide

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The review in one line

Noise goes on the controls; wrap every angle; floor every variance

Density inverts, sampler runs forward: test one against the other

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Where the book contradicts itself

- Table 5.5 lines 8–10 use δ̃; (5.44)–(5.46) on p. 139 and Table 5.6 use δ
- p. 132 says the sampler stays in [−2b, 2b]; the support is [−6b, 6b]
- Table 5.7 has two lines numbered 3 and an unbounded loop
- Book exercise 5's table numbers its last row 1

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Tracked robot: still untested

- Book says odometry is usually more accurate (pp. 121, 132); not tested on the hull
- T5 simulation log-likelihood per step: velocity 6.13, odometry 6.59, χ-corrected 7.06
- That slip model was chosen by the package; check it on logs
- Quadruped needs book exercise 7 (holonomic); drone needs dynamics

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